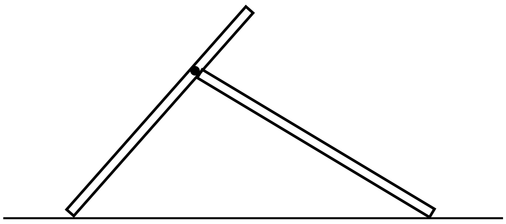
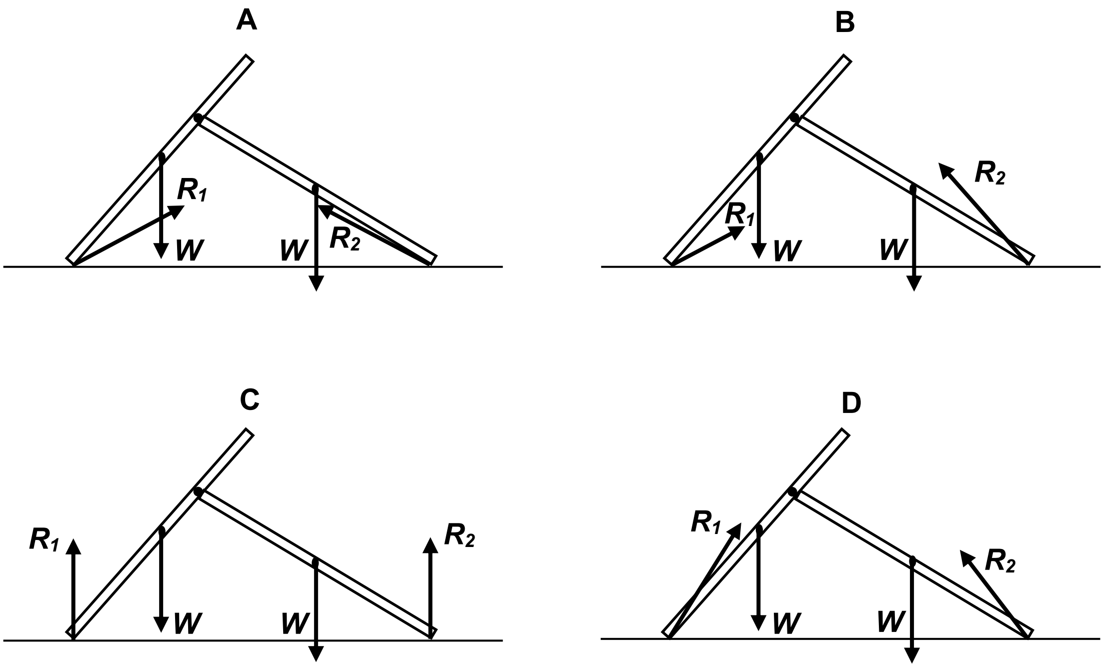
6 Two identical uniform rods, each of weight W, are hinged together to form a structure which is resting on a rough floor as shown.

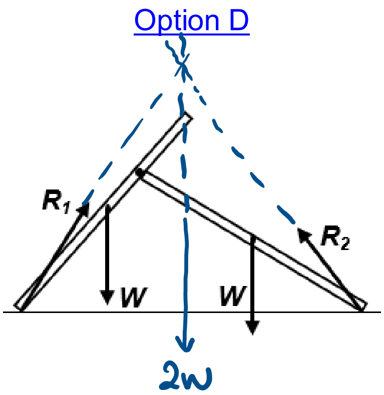


If the reaction forces acting on the structure by the floor are R_1 and R_2, which of the following shows the forces acting on the structure?



Ans: D

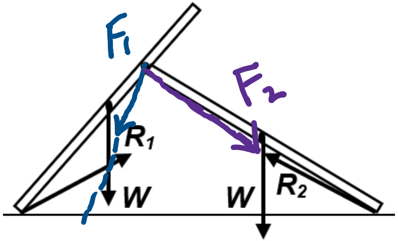
Combine the two W into a single $2W$, situated at the centre of the two W. Then extends the forces for $2W$ and the two R. There must intersect at one point. This will eliminate option B and C.



To eliminate option A, refer to the FBD of the left rod of A. There must be a force F_1 pointing downward and leftward to maintain equilibrium of the left rod.

Then, refer to FBD of right rod. There must be a F_2 pointing downward and rightward to maintain equilibrium of this rod. But as these two forces are action and reaction pair, they must point in opposite direction, which is not true in the drawing below. Hence option A is not the answer.

Option A



7 A student builds a launcher using an elastic cord of negligible mass and natural length of 1.0 m